Instructions: Thank you so much for your interest in WORQ! We hope this journey of discovery has been fruitful for you. We have certainly had a great time getting to know you. As part of that, we hope you may answer some questions to help us get to know you more! We invite you to add any information you wish that best showcases your abilities.

Using your own experience, please present your responses in a format that best showcases your **analytical thinking and process-building capability** (e.g., PowerPoint, Excel models, process maps, dashboards, or any format you deem suitable).

Thank you and we hope to hear from you soon!

Software Engineer Interview Questions

Please answer the questions below to the best of your ability with your own approach.

Expected time: 1-2 hours. Submit via a GitHub repo link with a working README.

Question 1 — REST API Design

Build a simple meeting room booking API with the following endpoints:

- POST /bookings — create a booking
- GET /bookings — list all bookings
- DELETE /bookings/:id — cancel a booking

Requirements:

- Use any programming language you are comfortable with.
- Bookings should have: room name, member name, start time, end time
- The API should reject a booking if the room is already taken for that time slot
- In-memory storage is fine, no need for a real database

Question 2 — Webhook Design

Extend your booking API to notify an external system whenever a booking is created or cancelled.

Requirements:

- Allow a consumer to register a webhook URL via POST /webhooks
- When a booking is created or cancelled, send a POST request to all registered webhook URLs with the relevant booking details
- If the webhook call fails, retry up to 3 times before giving up

---

Question 3 — Code Quality & Documentation

This applies to everything you submit above:

- A README with setup instructions and how to test each endpoint
- At least one unit test for the booking conflict logic
- Brief comments where the logic is non-obvious